VHDL Assignment 2

Subject: Digital Logic Design

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1. Basic Logic Gates (AND / OR)

1.1 VHDL Source code

```vhdl
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
entity logic_gate is
    port(
        a, b, c, d : in  std_logic;
        out1, out2 : out std_logic
    );
end entity logic_gate;

--architecture
architecture bhv of logic_gate is
begin
    out1 <= a and b and c and d;    -- 4-input AND
    out2 <= a or  b or  c or  d;    -- 4-input OR
end architecture bhv;
```

2. 8-to-1 Multiplexer

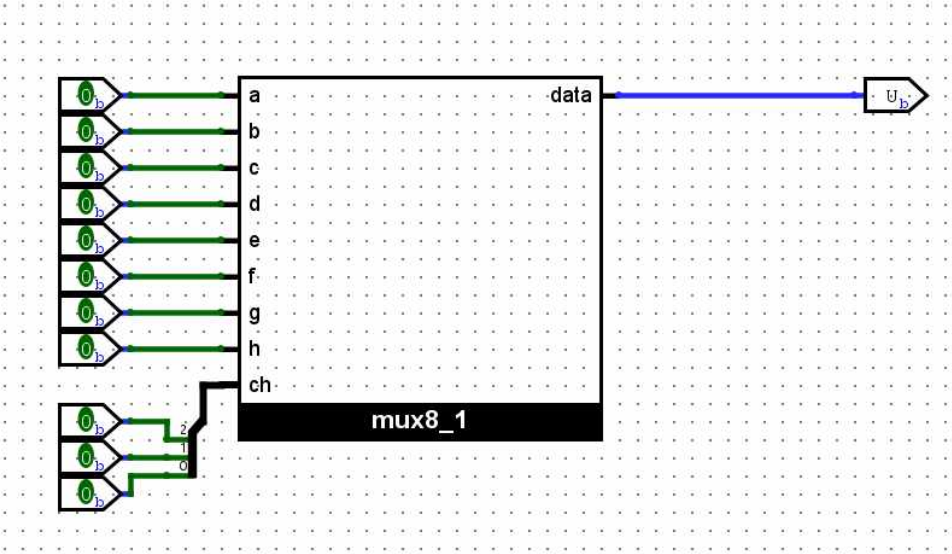
2.1 VHDL Source code

```vhdl
--library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

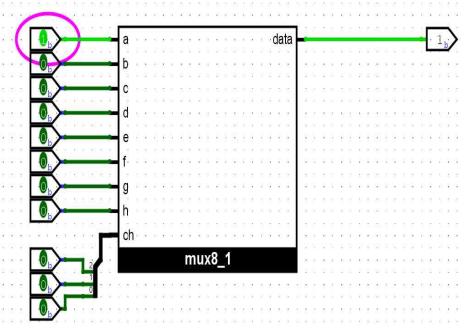
--entity
entity mux8_1 is
    port(
        a, b, c, d, e, f, g, h : in  std_logic;                    -- 입력 8개
        ch                     : in  std_logic_vector(2 downto 0); -- 채널 선택 1개
        data                   : out std_logic                     -- 출력 1개
    );
end entity mux8_1;

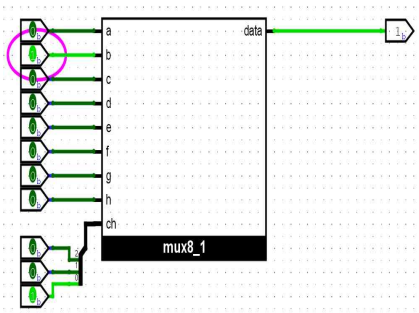
--architecture
architecture bhv of mux8_1 is
begin
    process(a, b, c, d, e, f, g, h, ch)
    begin
        case ch is
            when "000" => data <= a;
            when "001" => data <= b;
            when "010" => data <= c;
            when "011" => data <= d;
            when "100" => data <= e;
            when "101" => data <= f;
            when "110" => data <= g;
            when "111" => data <= h;
            when others => data <= '0';
        end case;
    end process;
end architecture bhv;
```

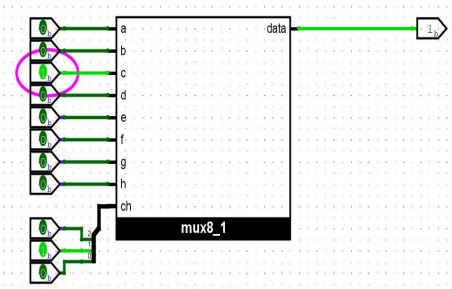
2.2 Circuit Configuration

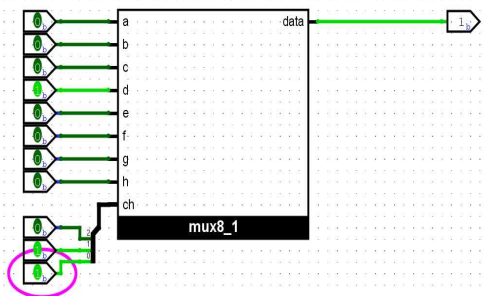


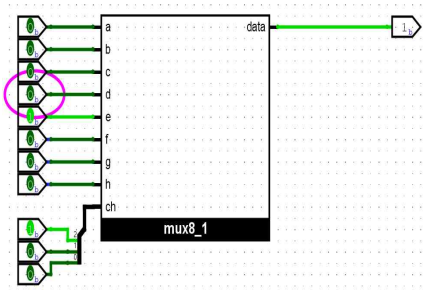
2.3 Simulation Results

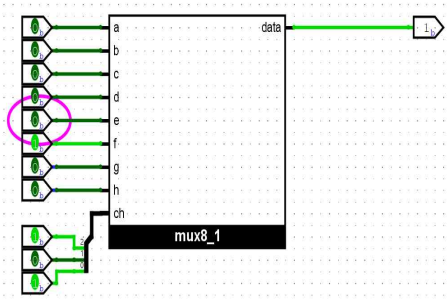


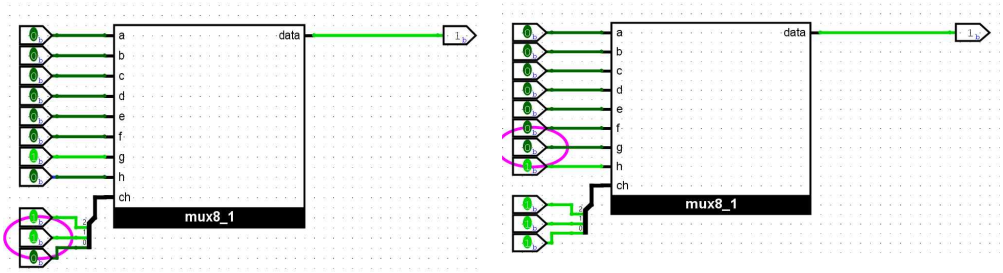




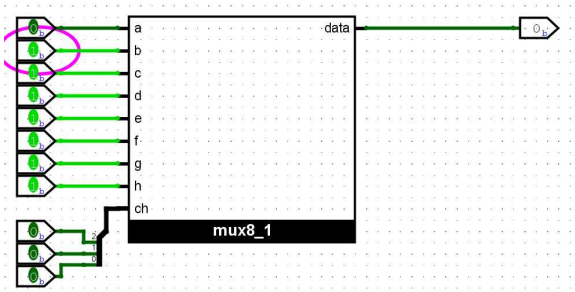








This circuit is an 8-to-1 multiplexer that uses a 3-bit channel input to select one of eight data inputs, so that only the selected input signal is passed to the output.



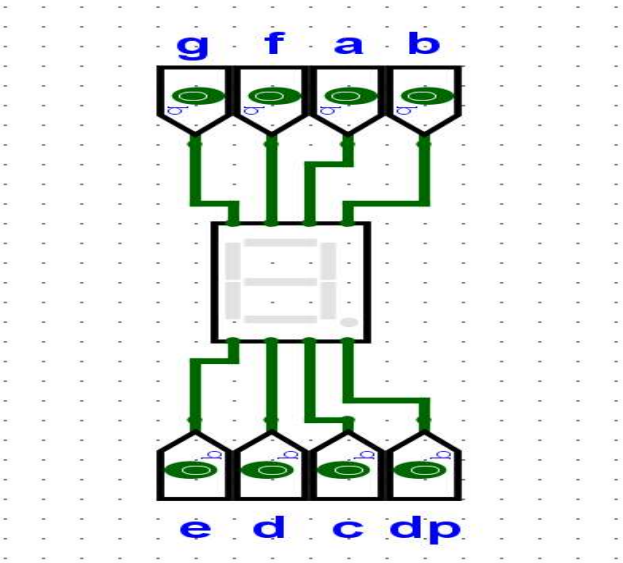
The output data follows only the selected input. Even if several data inputs are set to 1, the output stays at 0 unless the input chosen by the channel ch is 1.

ch2	ch1	ch0	selected input	data
0	0	0	a	a
0	0	1	b	b
0	1	0	c	c
0	1	1	d	d
1	0	0	e	e
1	0	1	f	f
1	1	0	g	g
1	1	1	h	h

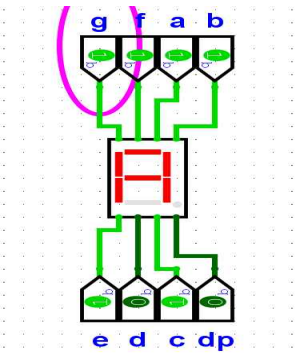
truth_table of 8 to 1 Multiplexer

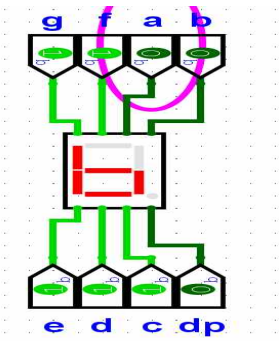
3. Letter Display on 7-Segment (A,B,C,D,F,G)

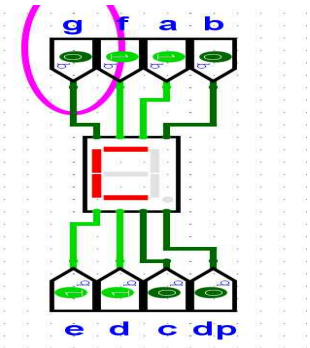
3.1 Circuit Configuration

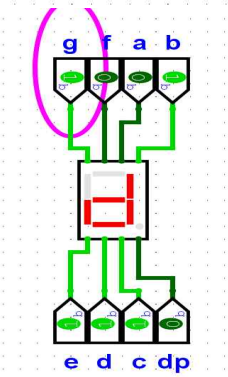


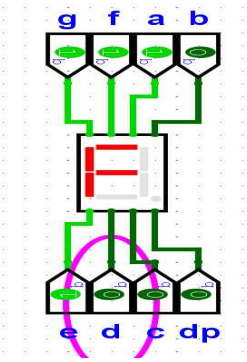
3.2 Simulation Results

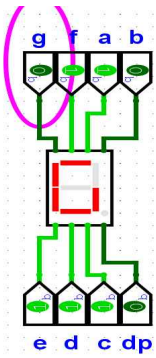












Letters b, d, and g look similar to other characters on a 7-segment display, so each letter must be drawn with its own lowercase segment pattern.

4. 25-Person Voting Machine

4.1 VHDL Source Code

```vhdl
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

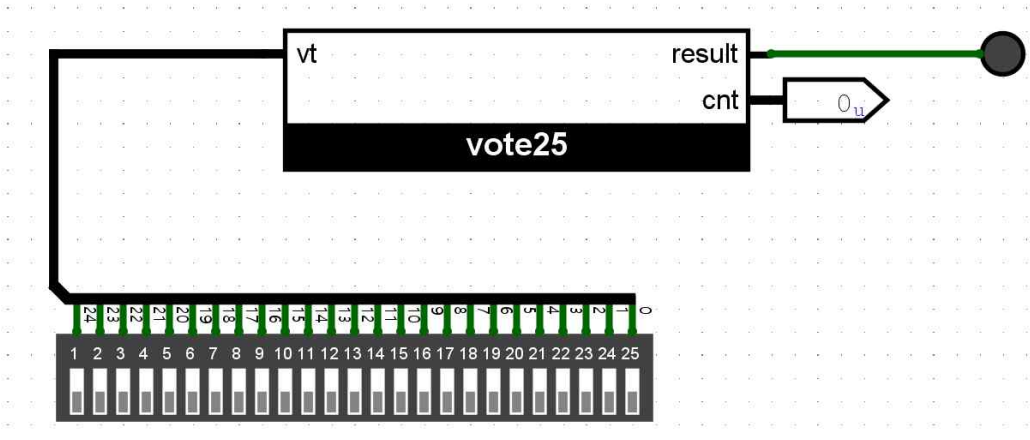
entity vote25 is
    port(
        vt     : in  std_logic_vector(25 downto 1);
        result : out std_logic;
        cnt    : out std_logic_vector(4 downto 0)   -- 진단용: 센 개수
    );
end entity vote25;

architecture bhv of vote25 is
begin
    process(vt)
        variable sum : integer range 0 to 25;
    begin
        sum := 0;
        for i in 1 to 25 loop
            if vt(i) = '1' then
                sum := sum + 1;
            end if;
        end loop;

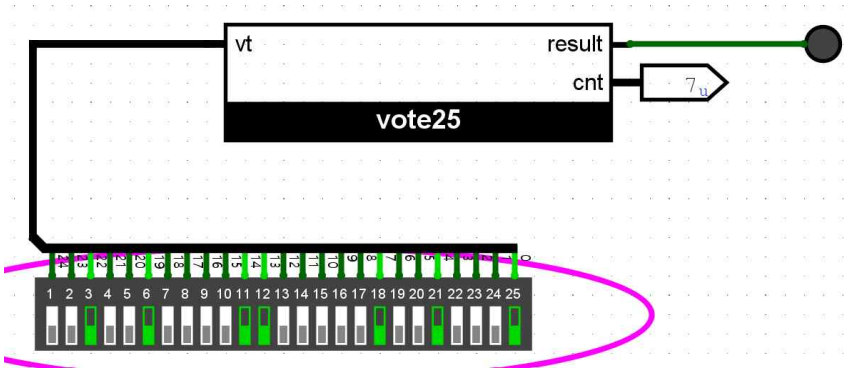
        cnt <= conv_std_logic_vector(sum, 5);   -- 개수를 5비트로 출력

        if sum >= 13 then
            result <= '1';
        else
            result <= '0';
        end if;
    end process;
end architecture bhv;
```

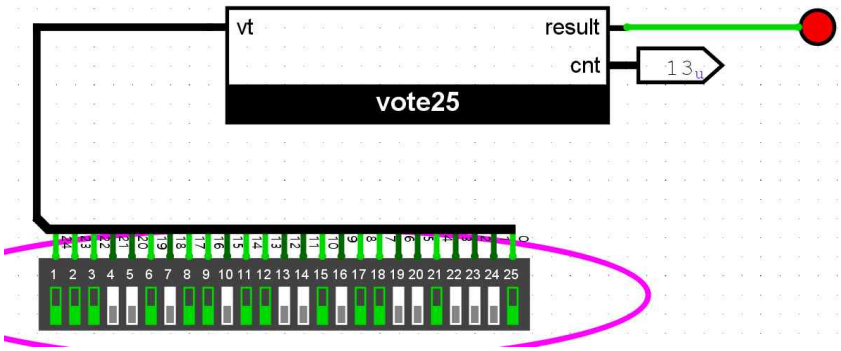
4.2 Circuit Configuration



4.3 Simulation Results



case 1) cnt(number of 1) < 13 , led turns off



case 2) cnt(number of 1) >= 13, led turns on

The circuit counts the number of inputs set to 1 among the 25 voting inputs, and the count is output as cnt. When cnt reaches 13 or more, which means a majority of the votes are 1, the LED turns on. Otherwise, the LED stays off.

5. One-Count Display with Enable

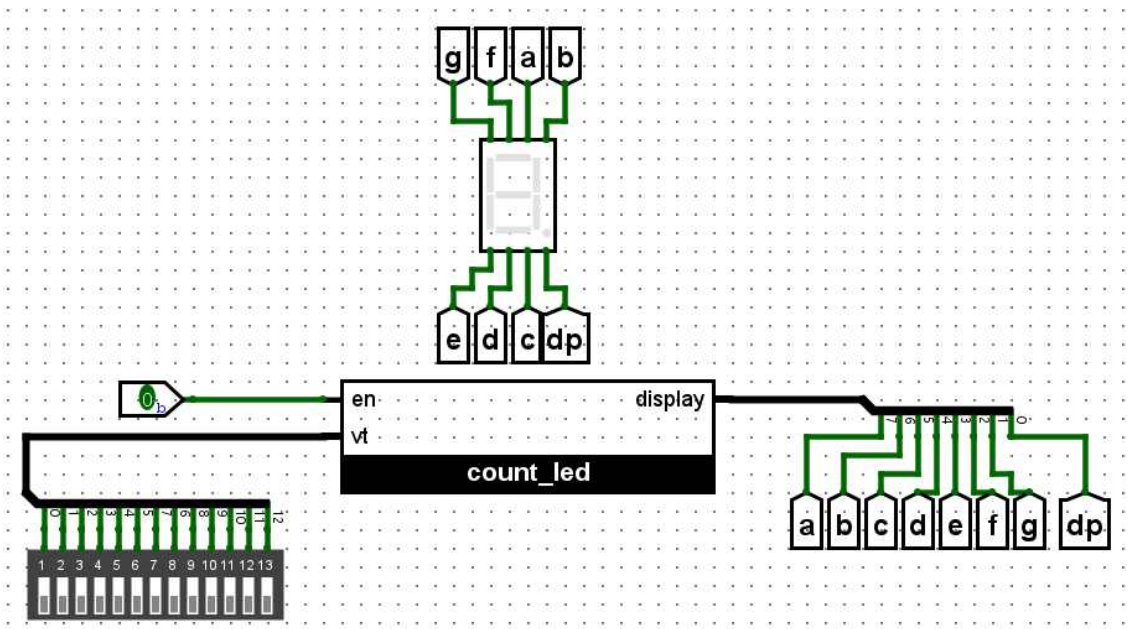
5.1 VHDL Source Code

```
1  --library and package
2  library ieee;
3  use ieee.std_logic_1164.all;
4  use ieee.std_logic_arith.all;
5  use ieee.std_logic_unsigned.all;
6
7  --entity
8  entity count_led is
9      port(
10         en      : in  std_logic;                    -- enable
11         vt      : in  std_logic_vector(13 downto 1);  -- 13비트 입력
12         display : out std_logic_vector(7 downto 0)    -- a b c d e f g dp
13     );
14 end entity count_led;
15
```

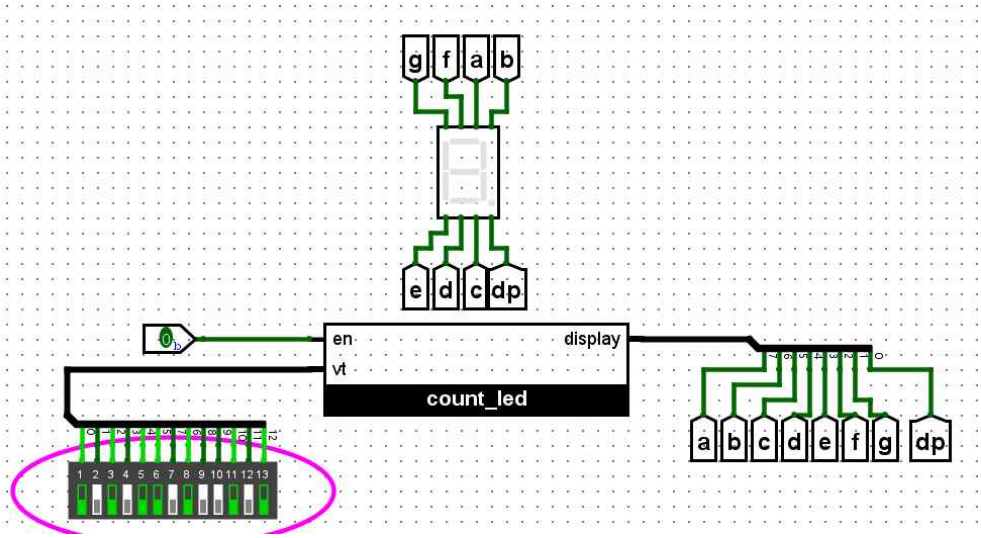
```vhdl
--architecture
architecture bhv of count_led is
begin
    process(en, vt)
        variable sum : integer range 0 to 13;
    begin
        -- 1. '1'의 개수 세기
        sum := 0;
        for i in 1 to 13 loop
            if vt(i) = '1' then
                sum := sum + 1;
            end if;
        end loop;

        -- 2. enable 판단 후 표시
        if en = '0' then
            display <= "00000000";                  -- 꺼짐
        else
            case sum is
                when 0  => display <= "11111100";   -- 0
                when 1  => display <= "01100000";   -- 1
                when 2  => display <= "11011010";   -- 2
                when 3  => display <= "11110010";   -- 3
                when 4  => display <= "01100110";   -- 4
                when 5  => display <= "10110110";   -- 5
                when 6  => display <= "10111110";   -- 6
                when 7  => display <= "11100000";   -- 7
                when 8  => display <= "11111110";   -- 8
                when 9  => display <= "11100110";   -- 9
                when 10 => display <= "11101110";   -- A
                when 11 => display <= "00111110";   -- b
                when 12 => display <= "10011100";   -- C
                when 13 => display <= "01111010";   -- d
                when others => display <= "00000000";
            end case;
        end if;
    end process;
end architecture bhv;
```

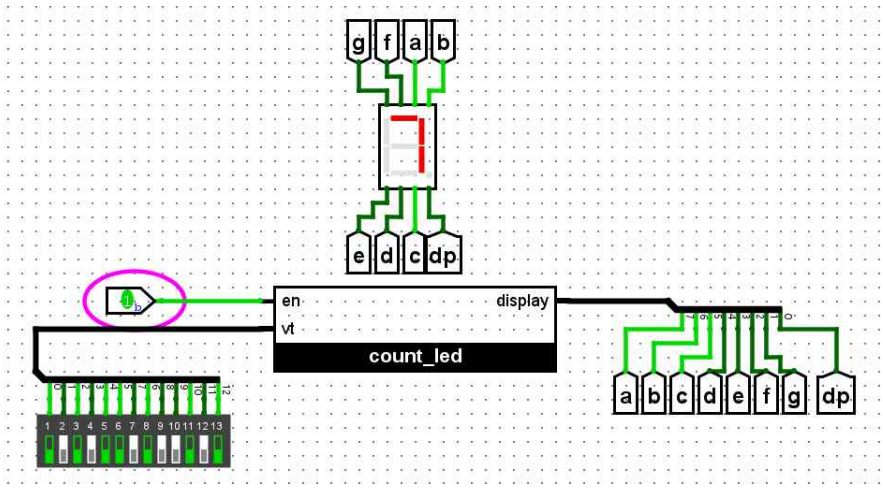
5.2 Circuit Configuration



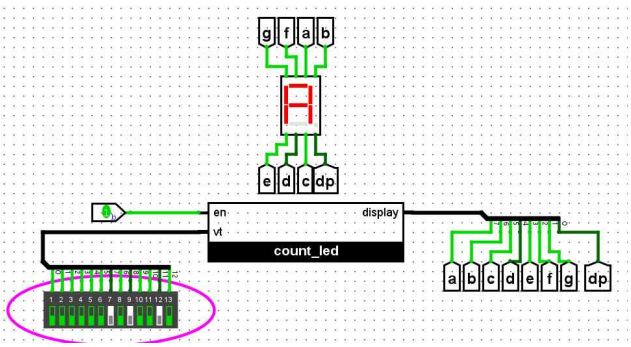
5.3 Simulation Results



case 1) en = 0, vt(number of 1's) = 7, circuit turns off



case 2) en = 1, vt = 7, circuit turns on, segment shows vt



case 3) en = 1, vt = 10 > 9 , segment shows the letter A instead of 10

When the enable signal en is set to 1, the circuit starts operating. It counts the number of 1s in the 13-bit input vt, and the result is shown on the 7-segment display as the corresponding digit or letter. When en is 0, the display is turned off.

If the count is between 0 and 9, the 7-segment display shows that value as a digit. If the count is between 10 and 15, the display shows the letters A, B, C, D, E, and F,

6. Conclusion

In this assignment, all five circuits were designed in VHDL and verified in simulation. The results matched the expected outputs for the tested inputs.